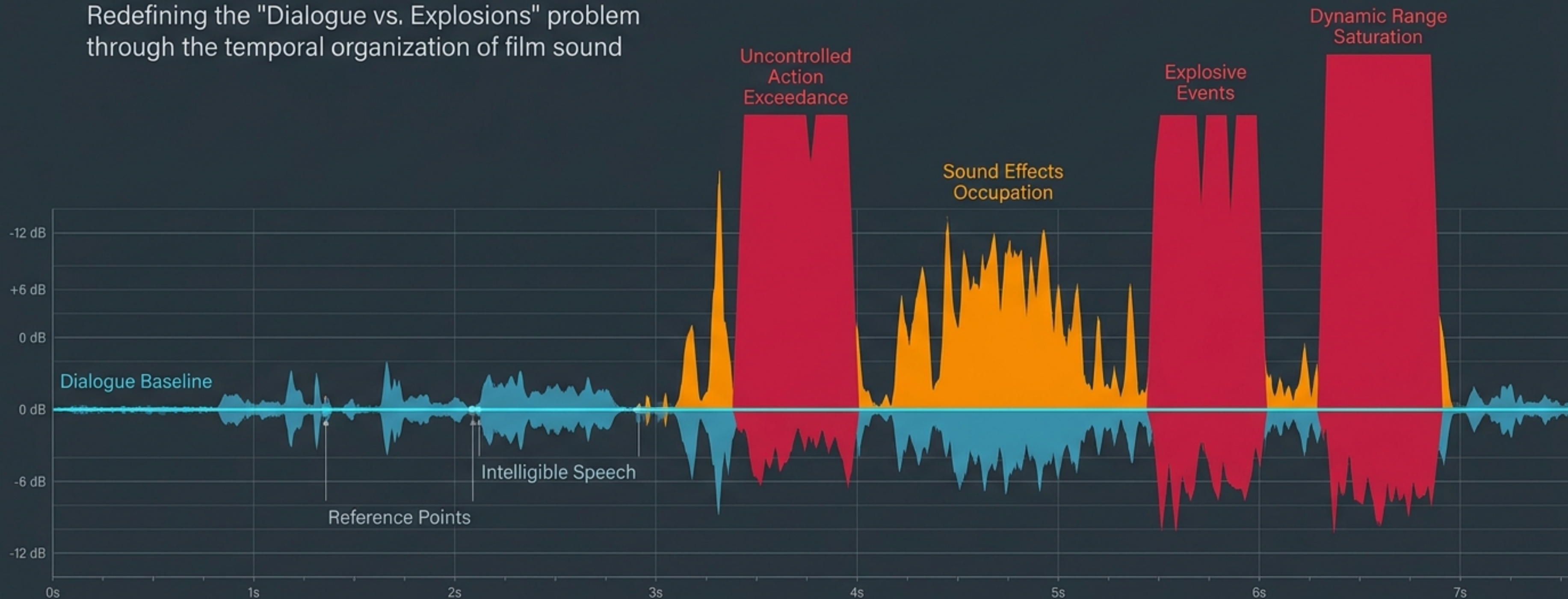
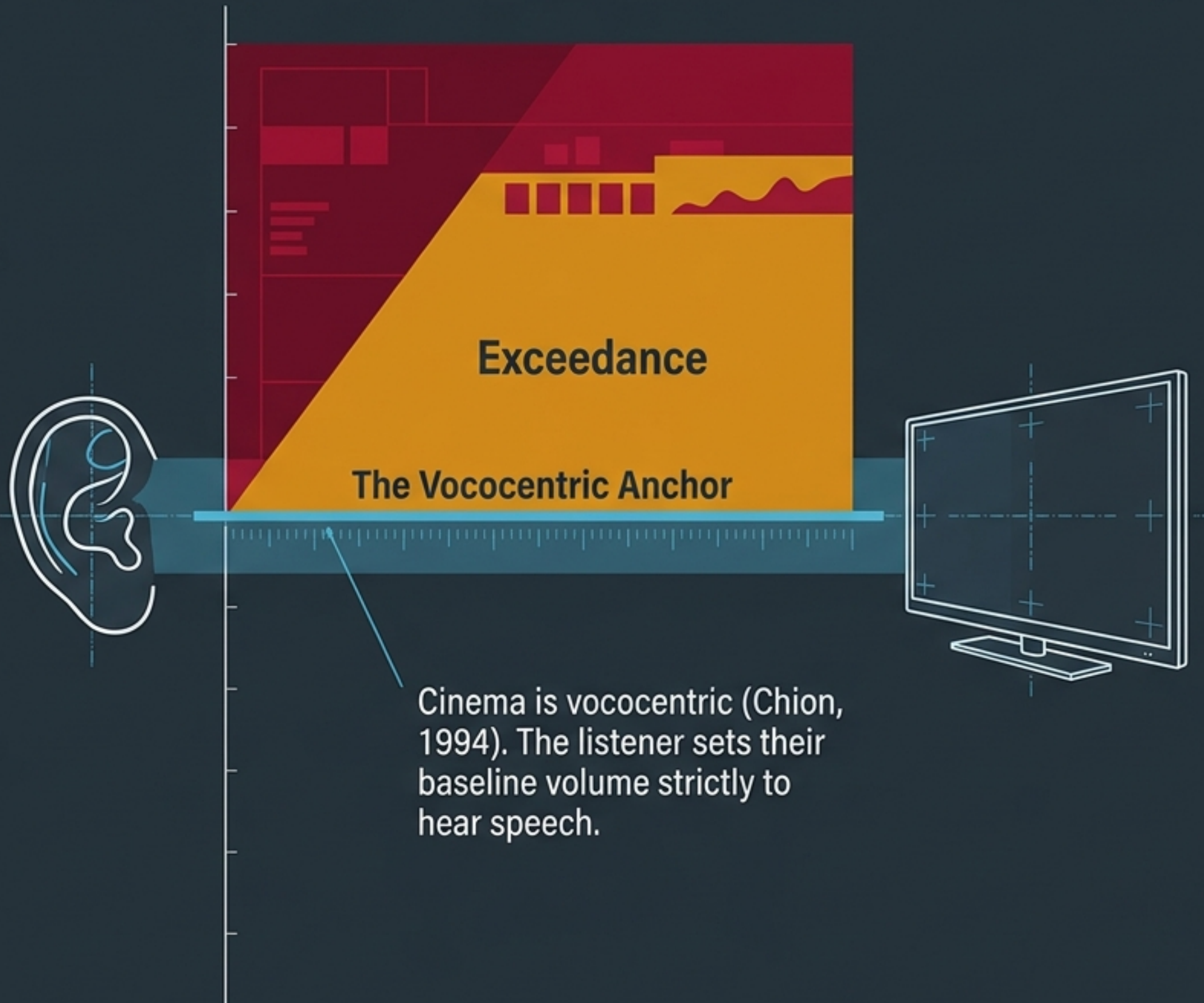


Acoustic Occupation

Redefining the "Dialogue vs. Explosions" problem through the temporal organization of film sound



A high-fidelity conceptual framework & case study



The Diagnostic Reality

The problem is not that action is loud, or that dynamic range is wide. Both are intended. The problem is three conditions occurring simultaneously:

1. The volume is anchored to speech.
2. Non-dialogue material heavily exceeds this level.
3. The exceedance persists for long, continuous stretches with zero recovery time.

The consequence: The listener's volume control becomes a reactive, manual component of the soundtrack's reproduction.

	The Industry Standard (Loudness)	The Proposed Framework (Occupation)
Core Focus	Distributional	Temporal
Metrics	ITU-R BS.1770 / LUFS	Exceedance Blocks & Recovery Debt
Reference Anchor	Absolute decibel scale	Relative to internal dialogue level
The Blind Spot	Ignores duration and listener fatigue	Explicitly measures duration and recovery gaps

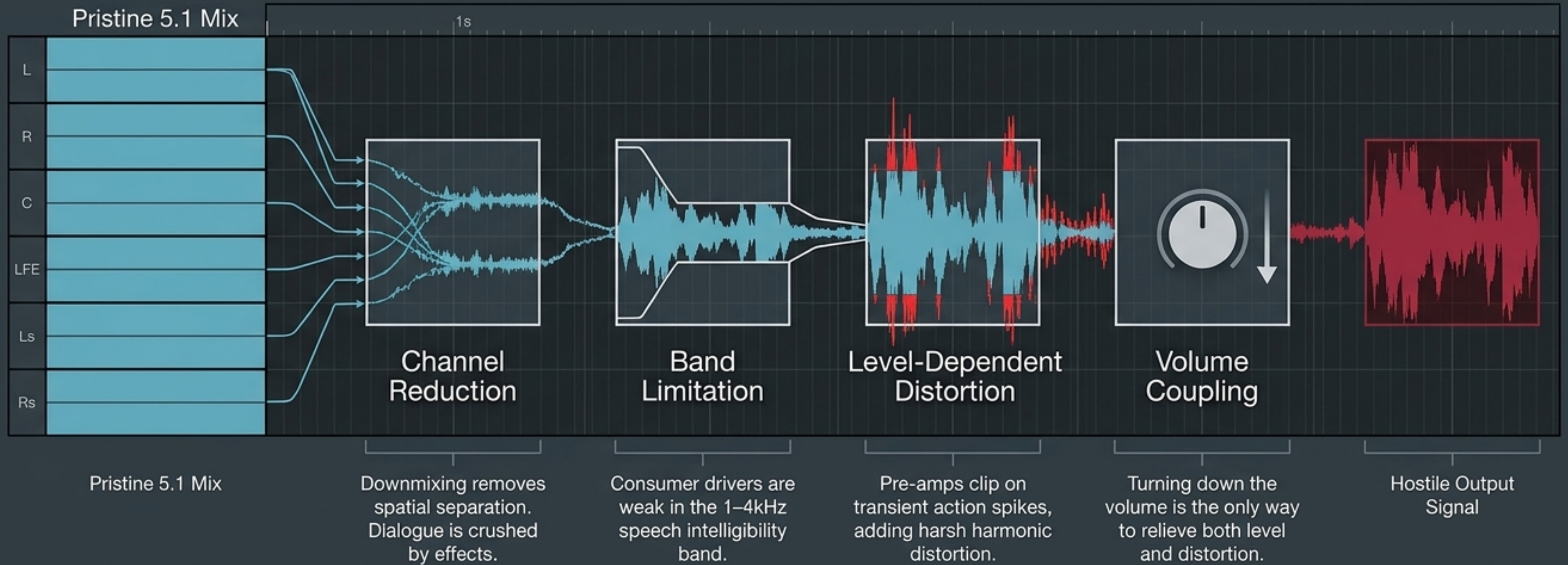
Integrated loudness summarizes time away into a single number.
Occupation maps exactly how sound occupies time.

The Reproduction Gap

[Claim: Established]

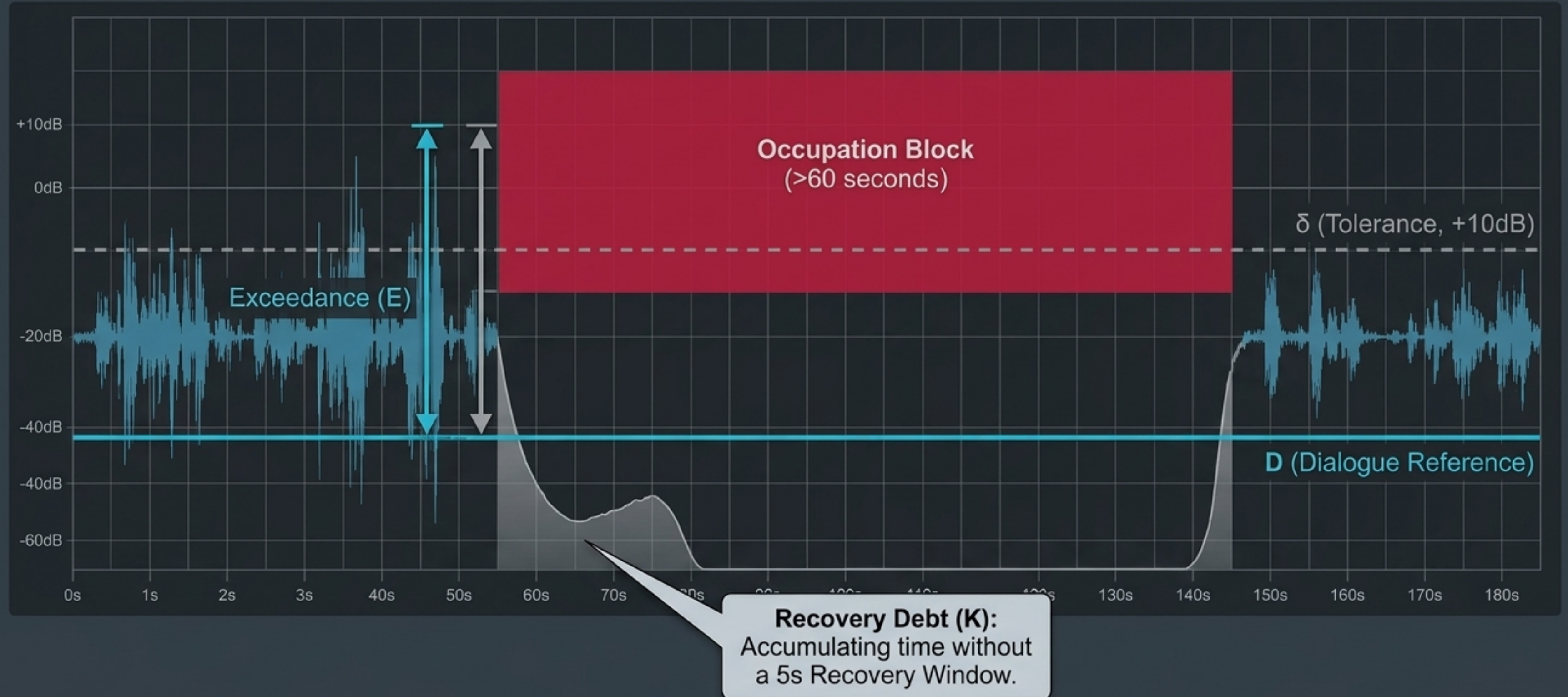
Calibrated Cinema (The Intent)	Uncalibrated Home (The Reality)
• Calibration: SMPTE standard 85 dB(C) reference. • Listener Anchor: System-fixed. • Spatial: Discrete 5.1/7.1 channels (speech separated from effects).	• Calibration: Arbitrary, uncalibrated room. • Listener Anchor: Set manually to speech intelligibility. • Spatial: Downmixed mono/stereo (effects crash into speech).

The Reproduction Chain Bottleneck



The Anatomy of an Occupation Block

Rendered in the 'Acoustic Dashboard'



Instrument Calibration & Failures

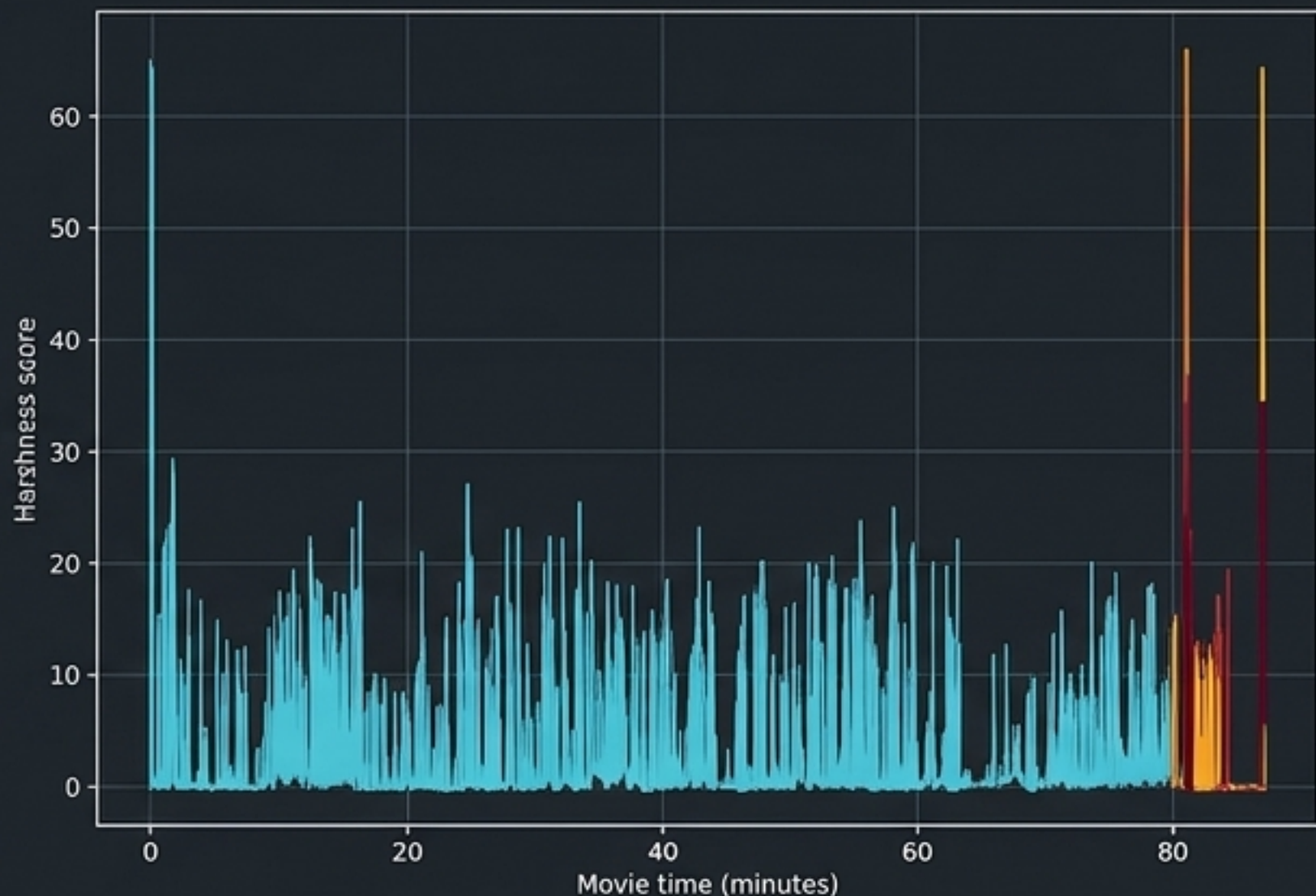
Panel 1: The False Swell



A zero MAD standard deviation inverted the onset trace. The detector found swells exactly where nothing happened.

Fix: Moved to a sustained 6dB event detector.

Panel 2: Harshness in Silence



Because spectral flatness is level-invariant, the digital noise floor of total silence triggered massive harshness scores.

Fix: Implemented strict audibility weighting.

A test that passes on synthetic data does not certify the code on real data. Both are required.

Case Study: Drive Through Fire (2026)

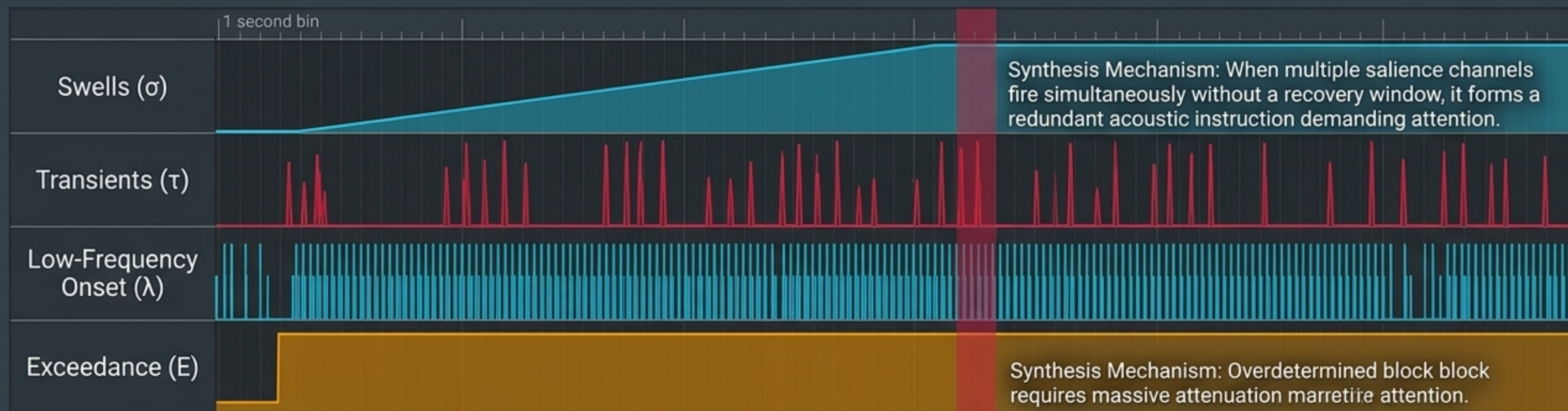
[Claim: Measured]



Frequent Return to Baseline.
(Median gap between recovery: 17.2s)

The 694.7-Second Gap.
For over 11.5 minutes, the soundtrack
never returns to its quieter states.

The Listener Model: Overdetermination



$$T_b = \max(0, Q95(E|b) - \delta)$$

Implied Throttle Depth. If the occupation block requires massive attenuation just to become tolerable, the listener is forced to mute the film entirely.

The Hand on the Volume: Behavioral Logging



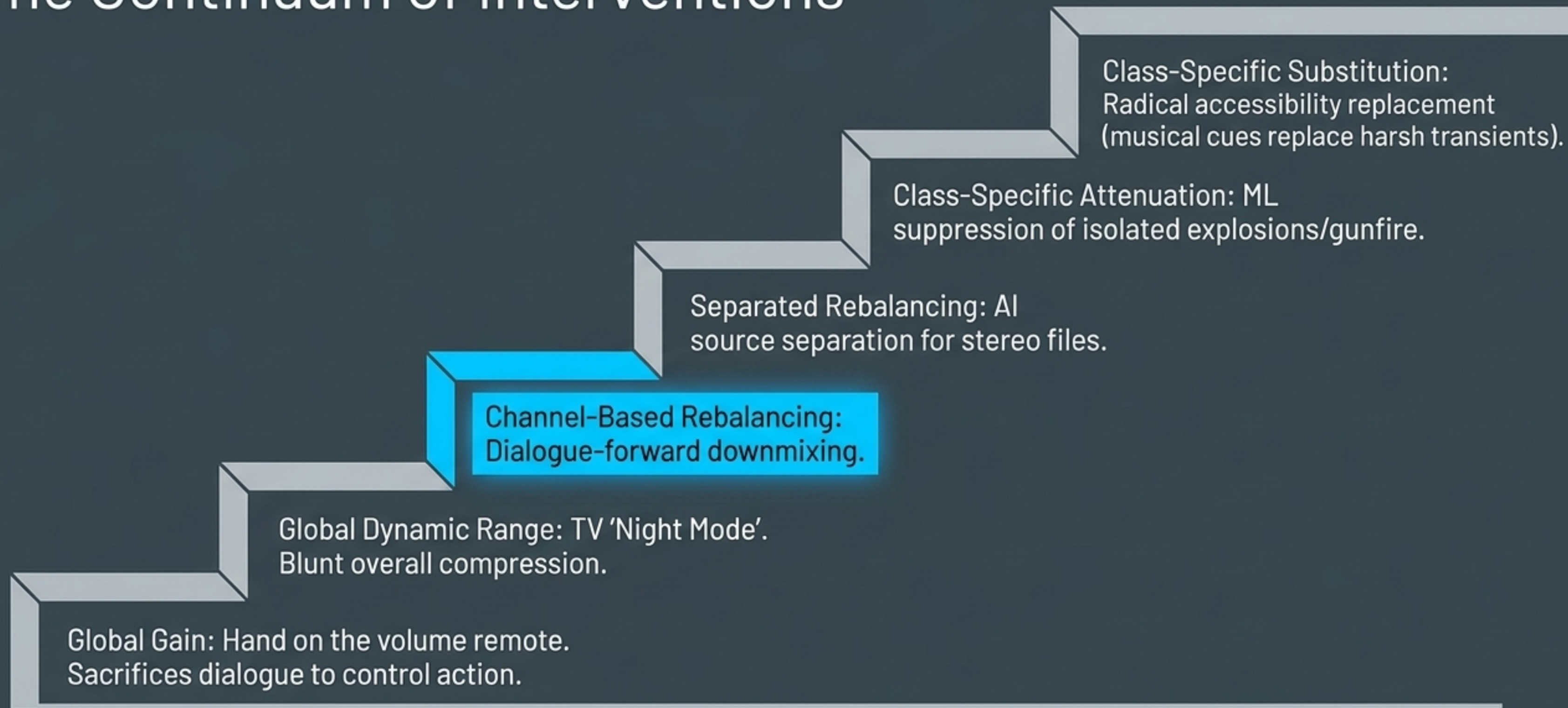
The Protocol (throttle-log.lua)

- Continuous variables: Reduction Onset, Reduction Depth, and Listener Recovery Time.
- Treats volume throttling as a measurable act of regaining control over a hostile soundscape (Glass and Singer, 1972).

The Hazard Model

Connects acoustic variables (Exceedance, Recovery Debt, Overdetermination) directly to the dependent behavioral variable: the probability of a volume reduction.

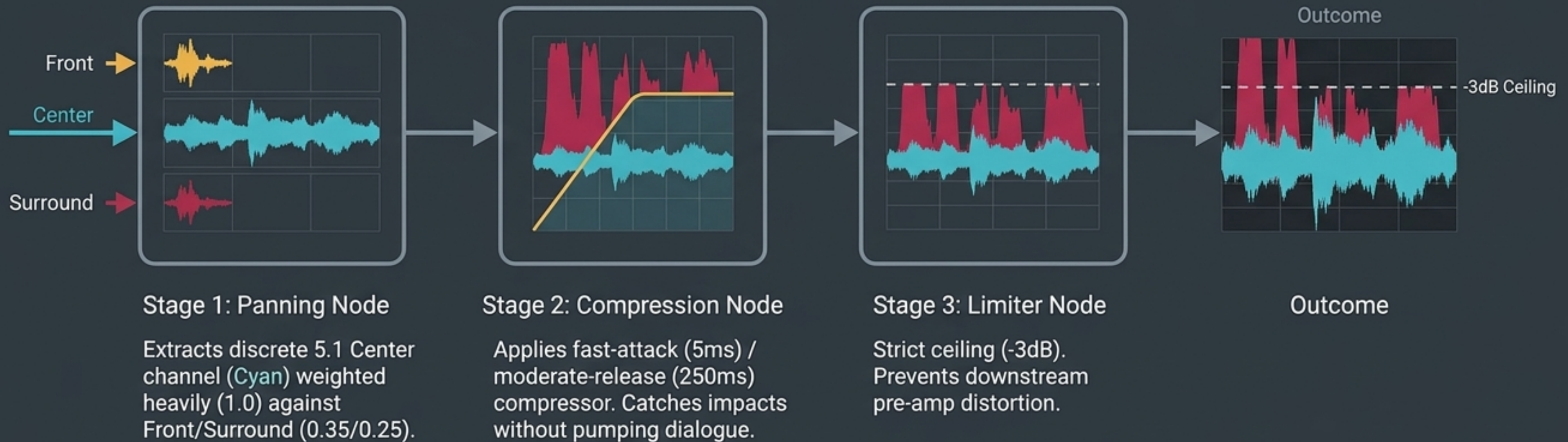
The Continuum of Interventions



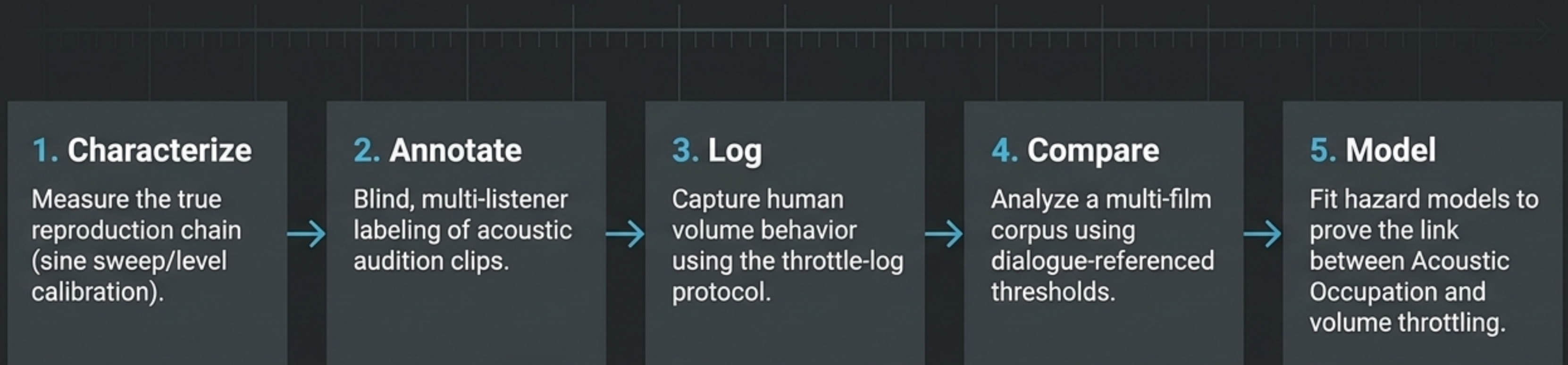
Crude / Consumer Intervention → Sophisticated / Systemic Intervention

The Dialogue-Forward Downmix

Acoustic Dashboard



The Research Programme



A complaint about a soundtrack is easy to confirm with plots drawn after the fact. It is harder, and more useful, to specify in advance what the soundtrack must contain to force the listener's hand.